Chapter 2: The Simple Regression Model: 2-1 Definition of the Simple Regression Model
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2-1 Definition of the Simple Regression Model

Much of applied econometric analysis begins with the following premise: *y* and *x* are two variables, representing some population, and we are interested in "explaining *y* in terms of *x*," or in "studying how *y* varies with changes in *x*." We discussed some examples in Chapter 1, including: *y* is soybean crop yield and *x* is amount of fertilizer; *y* is hourly wage and *x* is years of education; and *y* is a community crime rate and *x* is number of police officers.

In writing down a model that will "explain *y* in terms of *x*," we must confront three issues. First, because there is never an exact relationship between two variables, how do we allow for other factors to affect *y*? Second, what is the functional relationship between *y* and *x*? And third, how can we be sure we are capturing a ceteris paribus relationship between *y* and *x* (if that is a desired goal)?

We can resolve these ambiguities by writing down an equation relating *y* to *x*. A simple equation is

$$y = \beta_0 + \beta_1 x + u.$$

[2.1]

Equation (2.1), which is assumed to hold in the population of interest, defines the **simple linear regression model** (A model where the dependent variable is a linear function of a single independent variable, plus an error term.) . It is also called the *two-variable linear regression model* or *bivariate linear regression model* because it relates the two variables *x* and *y*. We now discuss the meaning of each of the quantities in equation (2.1). [Incidentally, the term "regression" has origins that are not especially important for most modern econometric applications, so we will not explain it here. See Stigler (1986) for an engaging history of regression analysis.]

When related by equation (2.1), the variables y and x have several different names used interchangeably, as follows: y is called the **dependent variable** (The variable to be explained in a multiple regression model (and a variety of other models).) , the **explained variable** (*See* dependent variable.) , the **response variable** (*See* dependent variable.) , the **predicted variable** (*See* dependent variable.) , or the **regressand** (*See* dependent variable.) ; x is called the **independent variable** (*See* explanatory variable.) , the **explanatory variable** (In regression analysis, a variable that is used to explain variation in the dependent variable.) , the **control variable** (*See* explanatory variable.) , the **predictor variable** (*See* explanatory variable.) , or the **regressor** (*See* explanatory variable.) . (The term **covariate** (*See* explanatory variable.) is also used for x.) The terms "dependent variable" and "independent variable" are frequently used in econometrics. But be aware that the label "independent" here does not refer to the statistical notion of independence between random variables (see Math Refresher B).

The terms "explained" and "explanatory" variables are probably the most descriptive. "Response" and "control" are used mostly in the experimental sciences, where the variable x is under the experimenter's control. We will not use the terms "predicted variable" and "predictor," although you sometimes see these in applications that are purely about prediction and not causality. Our terminology for simple regression is summarized in Table 2.1.

Table 2.1

Terminology for Simple Regression

Y	***X***
Dependent variable	Independent variable
Explained variable	Explanatory variable
Response variable	Control variable
Predicted variable	Predictor variable

Y	*X*
Regressand	Regressor

The variable *u*, called the **error term** (The variable in a simple or multiple regression equation that contains unobserved factors which affect the dependent variable. The error term may also include measurement errors in the observed dependent or independent variables.) or **disturbance** in the relationship, represents factors other than *x* that affect *y*. A simple regression analysis effectively treats all factors affecting *y* other than *x* as being unobserved. You can usefully think of *u* as standing for "unobserved."

Equation (2.1) also addresses the issue of the functional relationship between *y* and *x*. If the other factors in *u* are held fixed, so that the change in *u* is zero, $\Delta u = 0$, then *x* has a *linear* effect on *y*:

$$\Delta y = \beta_1 \Delta x$$

[2.2]

if

$$\Delta u = 0.$$

[2.2]

Thus, the change in *y* is simply β_1 multiplied by the change in *x*. This means that β_1 is the **slope parameter** (The coefficient on an independent variable in a multiple regression model.) in the relationship between *y* and *x*, holding the other factors in *u* fixed; it is of primary interest in applied economics. The **intercept parameter** (The parameter in a multiple linear regression model that gives the expected value of the dependent variable when all the independent variables equal zero.) β_0, sometimes called the *constant term*, also has its uses, although it is rarely central to an analysis.

Example 2.1

Soybean Yield and Fertilizer

Suppose that soybean yield is determined by the model

$$yield = \beta_0 + \beta_1 fertilizer + u,$$

[2.3]

so that $y = yield$ and $x = fertilizer$. The agricultural researcher is interested in the effect of fertilizer on yield, holding other factors fixed. This effect is given by β_1. The error term *u* contains factors such as land quality, rainfall, and so on. The coefficient β_1 measures the effect of fertilizer on yield, holding other factors fixed: $\Delta yield = \beta_1 \Delta\ fertilizer$.

Example 2.2

A Simple Wage Equation

A model relating a person's wage to observed education and other unobserved factors is

$$wage = \beta_0 + \beta_1 educ + u.$$

[2.4]

If *wage* is measured in dollars per hour and *educ* is years of education, then β_1 measures the change in hourly wage given another year of education, holding all other factors fixed. Some of those factors include labor force experience, innate ability, tenure with current employer, work ethic, and numerous other things.

The linearity of equation (2.1) implies that a one-unit change in *x* has the *same* effect on *y*, regardless of the initial value of *x*. This is unrealistic for many economic applications. For example, in the wage-education example, we might want to allow for *increasing* returns: the

next year of education has a *larger* effect on wages than did the previous year. We will see how to allow for such possibilities in Section 2-4.

The most difficult issue to address is whether model (2.1) really allows us to draw ceteris paribus conclusions about how *x* affects *y*. We just saw in equation (2.2) that β_1 *does* measure the effect of *x* on *y*, holding all other factors (in *u*) fixed. Is this the end of the causality issue? Unfortunately, no. How can we hope to learn in general about the ceteris paribus effect of *x* on *y*, holding other factors fixed, when we are ignoring all those other factors?

Section 2-5 will show that we are only able to get reliable estimators of β_0 and β_1 from a random sample of data when we make an assumption restricting how the unobservable *u* is related to the explanatory variable *x*. Without such a restriction, we will not be able to estimate the ceteris paribus effect, β_1. Because *u* and *x* are random variables, we need a concept grounded in probability.

Before we state the key assumption about how *x* and *u* are related, we can always make one assumption about *u*. As long as the intercept β_0 is included in the equation, nothing is lost by assuming that the average value of *u* in the population is zero. Mathematically,

$$\mathrm{E}(u) = 0.$$

[2.5]

Assumption (2.5) says nothing about the relationship between *u* and *x*, but simply makes a statement about the distribution of the unobserved factors in the population. Using the previous examples for illustration, we can see that assumption (2.5) is not very restrictive. In Example 2.1, we lose nothing by normalizing the unobserved factors affecting soybean yield, such as land quality, to have an average of zero in the population of all cultivated plots. The same is true of the unobserved factors in Example 2.2. Without loss of generality, we can assume that things such as average ability are zero in the population of all working people. If you are not convinced, you should work through Problem 2 to see that we can always redefine the intercept in equation (2.1) to make equation (2.5) true.

We now turn to the crucial assumption regarding how *u* and *x* are related. A natural measure of the association between two random variables is the *correlation coefficient*.

(See Math Refresher B for definition and properties.) If *u* and *x* are *uncorrelated*, then, as random variables, they are not *linearly* related. Assuming that *u* and *x* are uncorrelated goes a long way toward defining the sense in which *u* and *x* should be unrelated in equation (2.1). But it does not go far enough, because correlation measures only linear dependence between *u* and *x*. Correlation has a somewhat counterintuitive feature: it is possible for *u* to be uncorrelated with *x* while being correlated with functions of *x*, such as x^2. (See Section B-4 in Math Refresher B for further discussion.) This possibility is not acceptable for most regression purposes, as it causes problems for interpreting the model and for deriving statistical properties. A better assumption involves the *expected value of u given x*.

Because *u* and *x* are random variables, we can define the conditional distribution of *u* given any value of *x*. In particular, for any *x*, we can obtain the expected (or average) value of *u* for that slice of the population described by the value of *x*. The crucial assumption is that the average value of *u* does *not* depend on the value of *x*. We can write this assumption as

$$\mathrm{E}(u|x) = \mathrm{E}(u).$$

[2.6]

Equation (2.6) says that the average value of the unobservables is the same across all slices of the population determined by the value of *x* and that the common average is necessarily equal to the average of *u* over the entire population. When assumption (2.6) holds, we say that *u* is **mean independent** (The key requirement in multiple regression analysis, which says the unobserved error has a mean that does not change across subsets of the population defined by different values of the explanatory variables.) of *x*. (Of course, mean independence is implied by full independence between *u* and *x*, an assumption often used in basic probability and statistics.) When we combine mean independence with assumption (2.5), we obtain the **zero conditional mean assumption** (A key assumption used in multiple regression analysis that states that, given any values of the explanatory variables, the expected value of the error equals zero. (*See* Assumptions MLR.4, TS.3, and TS.3' in the text.)) , $\mathrm{E}(u|x) = 0$. It is critical to remember that equation (2.6) is the assumption with impact; assumption (2.5) essentially defines the intercept, β_0.

Let us see what equation (2.6) entails in the wage example. To simplify the discussion, assume that *u* is the same as innate ability. Then equation (2.6) requires that the average

level of ability is the same, regardless of years of education. For example, if $\mathbf{E}(abil|8)$ denotes the average ability for the group of all people with eight years of education, and $\mathbf{E}(abil|16)$ denotes the average ability among people in the population with sixteen years of education, then equation (2.6) implies that these must be the same. In fact, the average ability level must be the same for *all* education levels. If, for example, we think that average ability increases with years of education, then equation (2.6) is false. (This would happen if, on average, people with more ability choose to become more educated.) As we cannot observe innate ability, we have no way of knowing whether or not average ability is the same for all education levels. But this is an issue that we must address before relying on simple regression analysis.

In the fertilizer example, if fertilizer amounts are chosen independently of other features of the plots, then equation (2.6) will hold: the average land quality will not depend on the amount of fertilizer. However, if more fertilizer is put on the higher-quality plots of land, then the expected value of *u* changes with the level of fertilizer, and equation (2.6) fails.

The zero conditional mean assumption gives β_1 another interpretation that is often useful. Taking the expected value of equation (2.1) conditional on *x* and using $\mathbf{E}(u|x) = 0$ gives

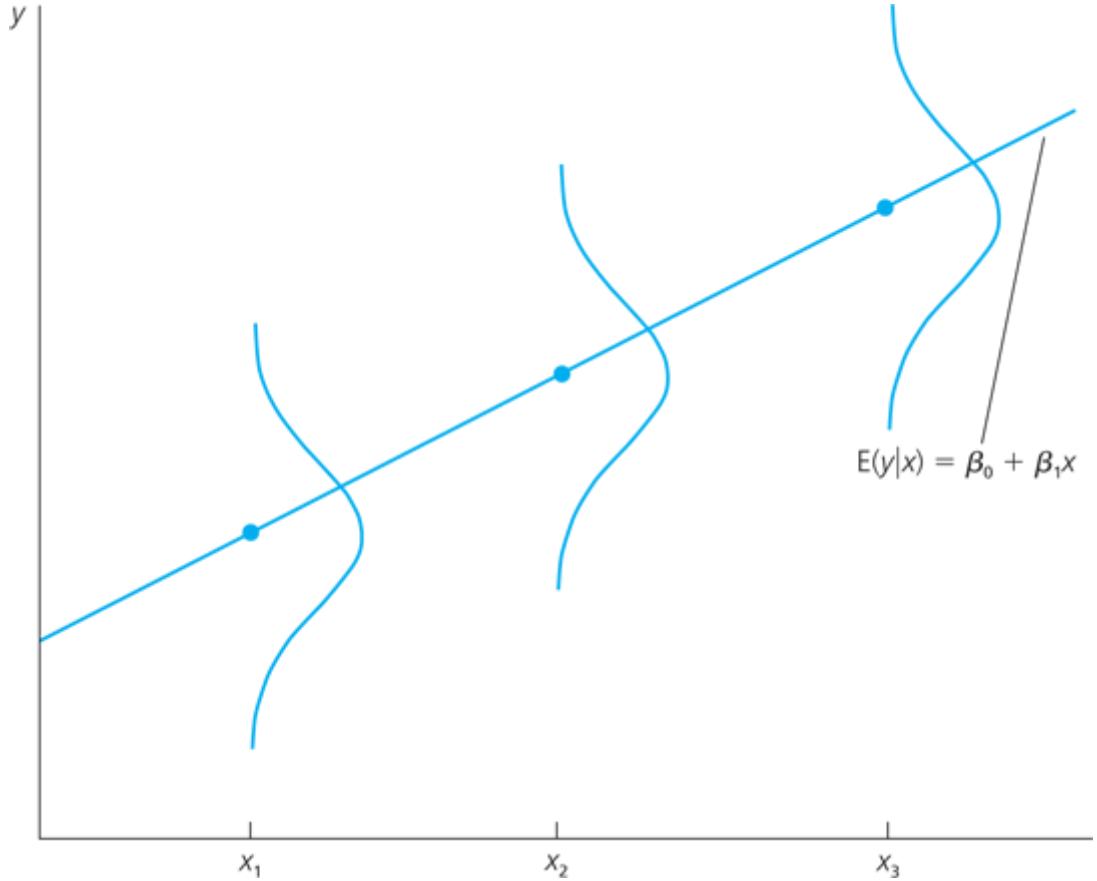
$$\mathbf{E}(y|x) = \beta_0 + \beta_1 x.$$

[2.8]

Equation (2.8) shows that the **population regression function (PRF)** (*See* conditional expectation.) , $\mathbf{E}(y|x)$, is a linear function of *x*. The linearity means that a one-unit increase in *x* changes the *expected value* of *y* by the amount β_1. For any given value of *x*, the distribution of *y* is centered about $\mathbf{E}(y|x)$, as illustrated in Figure 2.1.

Figure 2.1

$\mathbf{E}(y|x)$ as a linear function of *x*.



It is important to understand that equation (2.8) tells us how the *average* value of *y* changes with *x*; it does not say that *y* equals $\beta_0 + \beta_1 x$ for all units in the population. For example, suppose that *x* is the high school grade point average and *y* is the college GPA, and we happen to know that $\mathrm{E}(colGPA|hsGPA) = 1.5 + 0.5\,hsGPA$. [Of course, in practice, we never know the population intercept and slope, but it is useful to pretend momentarily that we do to understand the nature of equation (2.8).] This GPA equation tells us the *average* college GPA among all students who have a given high school GPA. So suppose that $hsGPA = 3.6$. Then the average *colGPA* for all high school graduates who attend college with $hsGPA = 3.6$ is $1.5 + 0.5(3.6) = 3.3$. We are certainly *not* saying that every student with $hsGPA = 3.6$ will have a 3.3 college GPA; this is clearly false. The PRF gives us a relationship between the average level of *y* at different levels of *x*. Some students with $hsGPA = 3.6$ will have a college GPA higher than 3.3, and some will have a lower college GPA. Whether the actual *colGPA* is above or below 3.3 depends on the unobserved factors

in *u*, and those differ among students even within the slice of the population with $hsGPA = 3.6$.

Given the zero conditional mean assumption $E(u|x) = 0$, it is useful to view equation (2.1) as breaking *y* into two components. The piece $\beta_0 + \beta_1 x$, which represents $E(y|x)$, is called the *systematic part* of *y*—that is, the part of *y* explained by *x*—and *u* is called the *unsystematic part*, or the part of *y* not explained by *x*. In Chapter 3, when we introduce more than one explanatory variable, we will discuss how to determine how large the systematic part is relative to the unsystematic part.

In the next section, we will use assumptions (2.5) and (2.6) to motivate estimators of β_0 and β_1 given a random sample of data. The zero conditional mean assumption also plays a crucial role in the statistical analysis in Section 2-5.

Going Further 2.1

Suppose that a score on a final exam, *score*, depends on classes attended (*attend*) and unobserved factors that affect exam performance (such as student ability). Then

$$score = \beta_0 + \beta_1 attend + u.$$

[2.7]

When would you expect this model to satisfy equation (2.6)?

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